WEAK ETALENESS IS EQUIVALENT TO AN ISOMORPHISM ON STRICT HENSELIZATIONS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 19109-19148, immediately following the statement at 19097-19107.

Editorial note. Seven cross-chapter pointers (Algebra, More on Algebra) plus one same-chapter lemma are recorded in assembly.external_references.

Editorial note. The statement's two conditions are retained verbatim in the statement span.

Allowed standard prerequisites (editorial).

- The stalkwise criterion for weak etaleness. chapter lemma lemma-check-weakly-etale-stalks, cited in the proof. Reduces the global statement to the local rings, which is what the strict henselization test is about.
- Permanence of weak etaleness under composition and base change. More on Algebra, lemmas more-algebra-lemma-weakly-etale-permanence and more-algebra-lemma-composition-weakly-etale, cited in the proof. Keeps weak etaleness along the ring maps that appear in the henselization diagram.
- Olivier's theorem on weakly etale maps and henselizations. More on Algebra, theorem more-algebra-theorem-olivier, cited at proof lines 19137-19138. Concludes that the comparison map of strict henselizations is an isomorphism.
- Functoriality of the strict henselization and go-down. Algebra, lemma algebra-lemma-strictly-henselian-functorial with More on Algebra, lemmas more-algebra-lemma-dumb-properties-henselization and more-algebra-lemma-go-down, cited in the proof. Controls the behaviour of strict henselizations along the maps of local rings.

Source reference. Original reference 04GU: focused-seeds/geometry/S9-morph-weakly-etale-strictly-henselian/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 094Q: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 18892–18892. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 092J: focused-seeds/geometry/S9-morph-weakly-etale-strictly-henselian/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original reference 07QM: focused-seeds/geometry/S9-morph-weakly-etale-strictly-henselian/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 092K: focused-seeds/geometry/S9-morph-weakly-etale-strictly-henselian/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 092L: focused-seeds/geometry/S9-morph-weakly-etale-strictly-henselian/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 092N: focused-seeds/geometry/S9-morph-weakly-etale-strictly-henselian/ref-tags.txt, source lines 10–10. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 092Z: focused-seeds/geometry/S9-morph-weakly-etale-strictly-henselian/ref-tags.txt, source lines 11–11. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of schemes. The following are equivalent*

1. *f is weakly étale, and*
2. *for $x \in X$ the local ring map $\mathcal{O}_{Y,f(x)} \to \mathcal{O}_{X,x}$ induces an isomorphism on strict henselizations.*

Proof. Let $x \in X$ be a point with image $y = f(x)$ in Y. Choose a separable algebraic closure κ^{sep} of $\kappa(x)$. Let $\mathcal{O}^{sh}_{X,x}$ be the strict henselization corresponding to κ^{sep} and $\mathcal{O}^{sh}_{Y,y}$ the strict henselization relative to the separable algebraic closure of $\kappa(y)$ in κ^{sep}. Consider the commutative diagram

$$\begin{array}{ccc} \mathcal{O}_{X,x} & \longrightarrow & \mathcal{O}^{sh}_{X,x} \\ \uparrow & & \uparrow \\ \mathcal{O}_{Y,y} & \longrightarrow & \mathcal{O}^{sh}_{Y,y} \end{array}$$

local homomorphisms of local rings, see Algebra, Lemma 04GU. Since the strict henselization is a filtered colimit of étale ring maps, More on Algebra, Lemma 092N shows the horizontal maps are weakly étale. Moreover, the horizontal maps are faithfully flat by More on Algebra, Lemma 07QM.

Assume f weakly étale. By Lemma 094Q the left vertical arrow is weakly étale. By More on Algebra, Lemmas 092J and 092L the right vertical arrow is weakly étale. By More on Algebra, Theorem 092Z we conclude the right vertical map is an isomorphism.

Assume $\mathcal{O}_{Y,y}^{sh} \to \mathcal{O}_{X,x}^{sh}$ is an isomorphism. Then $\mathcal{O}_{Y,y} \to \mathcal{O}_{X,x}^{sh}$ is weakly étale. Since $\mathcal{O}_{X,x} \to \mathcal{O}_{X,x}^{sh}$ is faithfully flat we conclude that $\mathcal{O}_{Y,y} \to \mathcal{O}_{X,x}$ is weakly étale by More on Algebra, Lemma 092K. Thus (2) implies (1) by Lemma 094Q. □